

Solution Architecture - High Level Design

Project Phoenix - AI-Powered Customer Service Platform

Author: Aisha Rahman & Priya Shah
Document version: 0.8 (draft)
Date: 3 June 2025

1. Component Overview

Component	AWS Service	Purpose
Voice channel	Amazon Connect	Inbound/outbound voice, call recording, IVR
Dialogue & intent mgmt	Amazon Lex	Coarse intent classification, slot filling for the top-8 trained intents
Generative / complex query handling	AWS Bedrock	Foundation model for intents outside the top-8, RAG over knowledge base
Knowledge base retrieval	Amazon OpenSearch + S3	Vector store for FAQ / policy content used in RAG
Orchestration	API Gateway + Lambda	Routes messages between Lex, Bedrock, and data layer
Data store	DynamoDB	Customer360, Orders, ConversationLog tables
Agent desktop	React (custom)	Human agent UI for handoffs and conversation history

2. Data Flow - Simple Intent

Customer message arrives via web widget or Connect → Lex classifies intent (one of top-8 trained intents, e.g. order status) → Lambda orchestrator performs slot filling → DynamoDB lookup (Orders / Customer360) → templated response returned to customer. Typical round trip target: under 1.5s.

3. Data Flow - Complex / Generative Intent

Customer message falls outside the top-8 trained intents → Lambda orchestrator retrieves relevant knowledge-base chunks from OpenSearch → conversation context + retrieved chunks passed to Bedrock → generated response returned to customer, with a confidence/sentiment check that may trigger escalation to a live agent via the /handoff endpoint. Target p95 round trip: 3.5s.

4. Non-Functional Requirements

- Performance: p95 response latency ≤ 3.5s for generative path, ≤ 1.5s for simple intent path.
- Availability: 99.5% uptime for customer-facing endpoints during contact-centre operating hours.
- Security: PII fields (email, address, payment references) encrypted at rest and in transit; access via least-privilege IAM roles.
- Data residency: all customer data stored within UK/EU AWS regions.
- Scalability: architecture must sustain Black-Friday-equivalent peak concurrency without manual intervention.

5. Open Items

Recommendation engine capability is not represented in this architecture and is not currently in scope (see Statement of Work, Section 3). Should this be approved via change control, an additional personalisation service and product-affinity data model would need to be scoped separately.